

FACTSHEET FS12 · Buy at the 52-week low

Should you buy stocks at their yearly low?

Henrik Gjerner · Rude Investment Consulting · Project 10 sidebar: strategy factsheets · data to 31 August 2026

STRATEGY Contrarian / bottom-fishing	ORIGIN Market folklore (bottom-fishing); tested against George & Hwang (2004)	FAMILY Reversal & bottom-fishing
UNIVERSES US · EU · UK · DK · SCANDI · World (point-in-time)	BACKTEST Feb 2013 – Aug 2026, monthly, net of costs	BOOKS Long/short and long-only (stocks closest to their 52-week low)
REBALANCE Monthly	STATUS Descriptive factsheet, not a registered trial	

The claim. Folklore says stocks at their yearly low are bargains: 'buy low'. The academic evidence points the other way: stocks near their lows tend to keep underperforming (George & Hwang 2004), and in the series A10 Free Fallin' found knives rebound only in market-wide crashes. **What we find, 2013–2026, point-in-time universes, after costs:** the long/short earned -17.0% to -5.2% a year and was positive in none of the six; no universe passes the 2.87 gate (significantly negative in US, EU, World). The long-only book (stocks closest to their 52-week low) had a higher Sharpe than the equal-weight universe in none of the six, with alphas of -9.7% to -2.3%, none past the gate. **What goes wrong (§10):** no single dominant drag. None of the pre-declared fixes passes the gate in both windows. **Classification (§11):** Reversal & bottom-fishing; its returns sit closest to the momentum cluster of the signal library. **360° view (§12):** nearest neighbours Momentum 12-1, Residual momentum; after those neighbours and the market, nothing is left (alpha $|t| < 2$ everywhere): the signal is redundant.

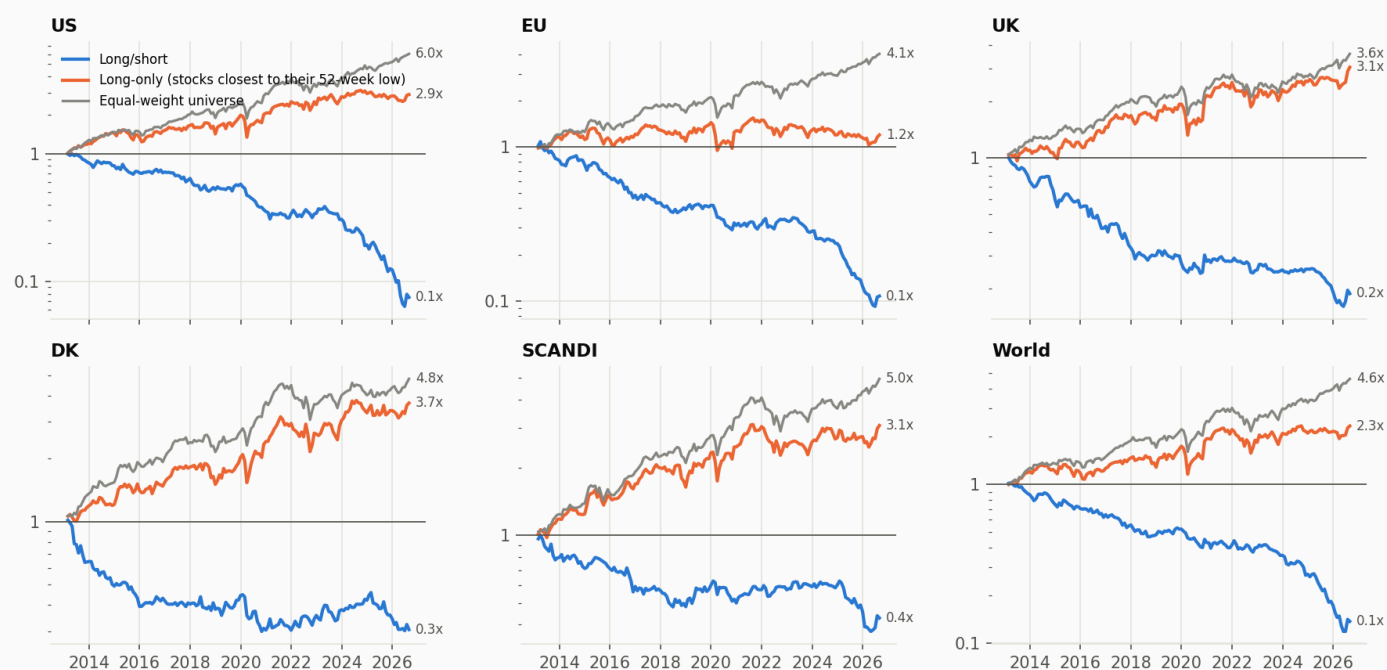
1. Headline performance

Universe	Names / groups	L/S return / yr	L/S t	L/S CAPM alpha (t)	L/S beta	Long- only CAGR	Long- only Sharpe	Long- only max DD	EW CAGR	EW Sharpe	Size- weighted CAGR
US	501 / 10	-17.0%	-3.40	-15.8% (-3.4)	-0.08	+8.1%	0.52	-33%	+14.1%	0.95	+15.3%
EU	238 / 10	-15.1%	-3.37	-16.6% (-3.7)	0.13	+1.4%	0.16	-34%	+10.9%	0.78	+10.1%
UK	239 / 10	-11.0%	-2.70	-14.0% (-3.6)	0.29	+8.6%	0.53	-32%	+9.9%	0.74	+8.5%
DK	19 / 3	-7.3%	-1.76	-8.5% (-1.9)	0.10	+10.1%	0.60	-32%	+12.3%	0.82	+10.3%

Universe	Names / groups	L/S return / yr	L/S t	L/S CAPM alpha (t)	L/S beta	Long-only CAGR	Long-only Sharpe	Long-only max DD	EW CAGR	EW Sharpe	Size-weighted CAGR
SCANDI	62 / 5	-5.2%	-1.49	-6.4% (-1.8)	0.09	+8.6%	0.58	-27%	+12.5%	0.91	+10.7%
World	971 / 10	-13.5%	-3.58	-15.0% (-4.4)	0.13	+6.4%	0.42	-33%	+11.9%	0.80	+10.8%

Monthly, local currency (World in USD). L/S = equal-weighted stocks closest to their 52-week low minus stocks furthest above it, net of 10 bp per side on turnover, a size-tiered borrow fee on the short leg and, where the book is net long cash, financing at the USD risk-free rate + 0.5%. Newey-West t (6 lags). CAPM vs the equal-weight universe. Size-weighted universe = market-cap weights in the US, traded-value weights elsewhere (a proxy for the index).

Buy at the 52-week low on six stock universes, 2013-2026



Growth of 1, log scale, local currency, net of costs. Henrik Gjering - Project 10 factsheets

2. Strategy description

Folklore says stocks at their yearly low are bargains: 'buy low'. The academic evidence points the other way: stocks near their lows tend to keep underperforming (George & Hwang 2004), and in the series A10 Free Fallin' found knives rebound only in market-wide crashes.

Why it might work. Folklore: mean reversion and value. Evidence: momentum and anchoring (George & Hwang 2004) predict continued weakness; losers carry high beta and volatility.

Rule	Original (Market folklore (bottom-fishing); tested against George & Hwang (2004))	This factsheet
Signal	Folklore: buy at or near the 52-week low	Total-return price / 252-day low; lowest = at the low

Rule	Original (Market folklore (bottom-fishing); tested against George & Hwang (2004))	This factsheet
Portfolio	No standard rule	Extreme quantile L/S (at the low minus far above), one month
Weighting	—	Equal weighted

3. Data load

Universe	Prices	Membership	Names eligible (median)	Currency
US	Sharadar SEP closeadj (total return), 805 tickers incl. delisted	Point-in-time: top 500 by market cap each month-end (Sharadar filings × price, A06 method)	~499	USD
EU	Project1 yfinance cache, Adj Close (TR)	Point-in-time: union of 11 national blue-chip indices (CAC 40, DAX, AEX, IBEX 35, FTSE MIB, OMXS30, OMXC25, OMXH25, BEL 20, PSI-20, WIG20), quarterly Wikipedia snapshots 2012–2026; 83% of member-quarters priced	~238	local (EUR, SEK, DKK, PLN)
UK	Project1 cache, Close + pence dividends (Yahoo under-adjusts LSE ~100×)	Point-in-time: FTSE 100 + FTSE 250, quarterly snapshots; 75% priced	~242	GBP
DK	Project1 cache, Adj Close	Point-in-time: OMX Copenhagen 25, quarterly snapshots; 91% priced	~19	DKK
SCANDI	Project1 cache, Adj Close	Point-in-time: OMXC25 + OMXS30 + OMXH25 (Oslo has no snapshot history); 88% priced	~62	local (DKK, SEK, EUR)
World	US Sharadar + UK + EU above	Point-in-time union of the three; local-currency returns summed, not converted	~963	mixed

Membership comes from Project2's cache of dated Wikipedia index pages (the same parsing as `run_wiki_pit.py` / `run_eu_pit.py`), with a short list of verified ticker renames (e.g. *NZYM-B* → *NSIS-B*, *WDH* → *DEMANT*, *DAI* → *MBG*). A name is eligible from the day after the snapshot that lists it, once it has 60 days of prices. Member-quarters without a price are mostly delisted names that Yahoo no longer serves, so some survivorship remains; see §13.

4. Signal creation

$LO52_{i,t} = P_{i,t} / \min(P_{i,t-251:t})$; the long leg holds the stocks with the lowest value (at the low). Signals use only information up to month-end t and are traded in month $t+1$. Eligibility: in the point-in-time universe on any of the last five trading days of month t .

5. Model build

- Sort eligible stocks into equal-count groups (deciles ≥ 100 names, quintiles 50–99, terciles < 50 ; fixed per universe).
- **Long/short** = stocks closest to their 52-week low minus stocks furthest above it; **long-only** = stocks closest to their 52-week low; benchmarks: equal-weight universe and size-weighted universe.
- Equal weights within each leg, rebalanced monthly.
- Costs: 10 bp per side on the one-way turnover of each leg; short-leg borrow fee 0.25% / 0.75% / 2% a year by size tier (largest 50% / next 30% / smallest 20% of the universe, `code/borrow.py`); net long cash financed at the USD risk-free rate + 0.5%. Code: `code/xs.py` (engine), `code/analyse_xs.py`, `code/build_xs.py`.

6. Performance in detail

US (top 500, point-in-time), median 501 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	-17.0%	-17.4%	19.5%	-0.87 [-1.35, -0.39]	-1.06	-94%	36%	-3.40	-0.08	-15.8% (-3.4)	14.1%
Long-only (net)	+9.5%	+8.1%	18.3%	0.52 [0.15, 0.98]	0.69	-33%	60%	2.86	1.08	-6.1% (-2.8)	10.7%
Stocks closest to their 52-week low (gross)	+10.6%	+9.3%	18.3%	0.58 [0.21, 1.05]	0.80	-32%	61%	3.21	1.08	-5.0% (-2.3)	10.6%
Stocks furthest above it (gross)	+25.2%	+25.4%	21.8%	1.15 [0.70, 1.62]	2.23	-20%	65%	4.90	1.16	+8.4% (2.7)	10.9%
EW universe	+14.4%	+14.1%	15.1%	0.95 [0.52, 1.53]	1.52	-25%	68%	4.79	1.00	–	8.9%
Size-weighted universe	+15.4%	+15.3%	14.4%	1.07 [0.61, 1.64]	1.79	-24%	69%	5.27	0.90	+2.4% (1.8)	8.3%

EU (11 national blue-chip indices, point-in-time), median 238 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	-15.1%	-15.2%	15.7%	-0.96 [-1.61, -0.40]	-1.13	-92%	37%	-3.37	0.13	-16.6% (-3.7)	9.6%

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
Long-only (net)	+2.9%	+1.4%	17.9%	0.16 [-0.29, 0.64]	0.11	-34%	52%	0.69	1.11	-9.7% (-4.8)	10.3%
Stocks closest to their 52-week low (gross)	+4.1%	+2.5%	17.9%	0.23 [-0.22, 0.70]	0.21	-34%	52%	0.97	1.11	-8.6% (-4.2)	10.2%
Stocks furthest above it (gross)	+16.7%	+16.3%	17.4%	0.96 [0.42, 1.54]	1.65	-35%	63%	3.45	0.98	+5.5% (1.8)	8.7%
EW universe	+11.4%	+10.9%	14.5%	0.78 [0.33, 1.33]	1.22	-26%	64%	3.34	1.00	–	8.3%
Size-weighted universe	+10.7%	+10.1%	14.3%	0.75 [0.28, 1.28]	1.14	-26%	60%	3.12	0.95	-0.2% (-0.1)	8.3%

UK (FTSE 350, point-in-time), median 239 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	-11.0%	-11.6%	16.4%	-0.67 [-1.21, -0.19]	-0.93	-84%	38%	-2.70	0.29	-14.0% (-3.6)	9.8%
Long-only (net)	+10.0%	+8.6%	18.9%	0.53 [0.12, 0.97]	0.78	-32%	56%	2.35	1.18	-2.3% (-1.1)	9.7%
Stocks closest to their 52-week low (gross)	+11.1%	+9.8%	18.9%	0.59 [0.18, 1.03]	0.92	-32%	57%	2.64	1.18	-1.2% (-0.6)	9.6%
Stocks furthest above it (gross)	+19.7%	+20.1%	16.0%	1.23 [0.71, 1.85]	2.19	-25%	64%	4.97	0.88	+10.4% (4.0)	9.0%
EW universe	+10.5%	+9.9%	14.1%	0.74 [0.26, 1.34]	1.08	-29%	60%	3.23	1.00	–	8.8%
Size-weighted universe	+9.0%	+8.5%	12.7%	0.71 [0.21, 1.31]	1.01	-27%	62%	2.94	0.82	+0.4% (0.2)	7.7%

DK (OMXC25, point-in-time), median 19 names, 3 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	-7.3%	-8.4%	16.9%	-0.43 [-0.89, 0.06]	-0.62	-71%	46%	-1.76	0.10	-8.5% (-1.9)	11.9%
Long-only (net)	+11.5%	+10.1%	19.1%	0.60 [0.17, 1.10]	0.82	-32%	60%	2.68	1.08	-2.4% (-1.1)	11.6%
Stocks closest to their 52-week low (gross)	+12.0%	+10.7%	19.1%	0.63 [0.21, 1.13]	0.88	-31%	60%	2.82	1.08	-1.8% (-0.9)	11.5%
Stocks furthest above it (gross)	+17.6%	+17.2%	18.3%	0.96 [0.36, 1.63]	1.70	-35%	63%	3.27	0.98	+5.0% (1.8)	9.2%
EW universe	+12.8%	+12.3%	15.6%	0.82 [0.30, 1.44]	1.25	-33%	64%	3.20	1.00	—	9.3%
Size-weighted universe	+11.2%	+10.3%	16.6%	0.68 [0.11, 1.32]	0.94	-40%	61%	2.52	0.92	-0.5% (-0.2)	10.8%

SCANDI (OMXC25 + OMXS30 + OMXH25, point-in-time), median 62 names, 5 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	-5.2%	-6.0%	14.3%	-0.36 [-0.81, 0.10]	-0.55	-63%	44%	-1.49	0.09	-6.4% (-1.8)	9.5%
Long-only (net)	+9.7%	+8.6%	16.9%	0.58 [0.12, 1.09]	0.81	-27%	58%	2.56	1.06	-3.9% (-2.0)	9.6%
Stocks closest to their 52-week low (gross)	+10.5%	+9.4%	16.9%	0.62 [0.17, 1.14]	0.90	-26%	58%	2.77	1.06	-3.1% (-1.6)	9.6%
Stocks furthest above it (gross)	+13.5%	+12.9%	16.2%	0.83 [0.28, 1.42]	1.37	-34%	58%	3.16	0.96	+1.1% (0.5)	8.3%
EW universe	+12.8%	+12.5%	14.1%	0.91 [0.39, 1.54]	1.42	-29%	67%	3.62	1.00	—	8.3%
Size-weighted universe	+11.2%	+10.7%	13.9%	0.80 [0.31, 1.37]	1.21	-27%	63%	3.26	0.95	-1.0% (-0.9)	8.6%

World (US + UK + EU, point-in-time), median 971 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	-13.5%	-13.6%	14.7%	-0.91 [-1.46, -0.42]	-1.12	-88%	39%	-3.58	0.13	-15.0% (-4.4)	9.5%
Long-only (net)	+8.1%	+6.4%	19.3%	0.42 [0.00, 0.87]	0.53	-33%	58%	1.93	1.14	-6.3% (-3.8)	10.8%
Stocks closest to their 52-week low (gross)	+9.2%	+7.6%	19.3%	0.48 [0.07, 0.93]	0.64	-33%	58%	2.22	1.14	-5.1% (-3.1)	10.7%
Stocks furthest above it (gross)	+20.2%	+20.2%	18.3%	1.10 [0.61, 1.67]	1.96	-27%	64%	4.45	1.02	+7.5% (3.4)	10.1%
EW universe	+12.5%	+11.9%	15.7%	0.80 [0.33, 1.38]	1.20	-29%	64%	3.53	1.00	–	9.4%
Size-weighted universe	+11.4%	+10.8%	15.0%	0.76 [0.29, 1.33]	1.14	-28%	63%	3.23	0.94	-0.4% (-0.5)	9.0%

Sharpe CI: stationary bootstrap (mean block 6 months, 5,000 draws). Max drawdown on monthly returns.

Calendar-year returns, L/S (net)

Year	US	EU	UK	DK	SCANDI	World
2013	-15.6%	-18.3%	-25.4%	-35.5%	-20.3%	-14.0%
2014	-4.9%	-6.0%	-20.2%	-23.7%	-10.8%	-13.1%
2015	-12.0%	-17.4%	-8.8%	-20.0%	-0.1%	-5.7%
2016	+1.6%	-23.4%	-20.9%	+6.3%	-16.7%	-8.4%
2017	-10.9%	-10.8%	-23.6%	-3.5%	-4.2%	-13.9%
2018	-18.2%	-9.8%	-8.6%	-5.5%	-15.1%	-13.2%
2019	+10.4%	+6.9%	-9.5%	+16.4%	+24.8%	+8.6%
2020	-39.0%	-26.1%	+2.6%	-31.7%	-16.8%	-17.8%
2021	-3.6%	+1.9%	+0.2%	-0.1%	+14.5%	-3.4%
2022	+0.7%	+5.2%	-10.2%	+17.7%	-0.1%	-3.6%
2023	-10.4%	-14.4%	-0.0%	+6.4%	+3.3%	-10.7%
2024	-36.9%	-17.0%	+1.0%	+11.0%	+1.8%	-24.0%
2025	-36.1%	-47.4%	-24.4%	-18.3%	-23.7%	-36.1%
2026	-39.2%	-13.5%	-2.3%	-11.3%	-6.5%	-21.0%

Calendar-year returns, long-only (net)

Year	US	EU	UK	DK	SCANDI	World
2013	+20.5%	+15.8%	+9.8%	+19.1%	+18.0%	+20.6%
2014	+19.8%	-3.2%	-7.9%	+1.5%	+5.2%	+0.0%
2015	-7.5%	-0.6%	+15.8%	+16.4%	+18.4%	-2.4%
2016	+13.7%	+9.2%	+24.9%	+10.0%	+15.0%	+7.3%
2017	+9.5%	+6.3%	+15.0%	+17.9%	+17.7%	+16.3%
2018	-14.4%	-15.6%	-5.8%	-17.4%	-17.8%	-12.8%
2019	+40.2%	+32.6%	+22.1%	+35.0%	+42.3%	+35.9%
2020	+3.8%	-11.5%	+11.3%	+18.8%	+6.1%	+11.4%
2021	+22.8%	+17.4%	+17.3%	+22.0%	+23.1%	+14.3%
2022	-5.2%	-13.3%	-16.5%	-11.1%	-18.5%	-16.9%
2023	+15.6%	-0.8%	+9.8%	+16.4%	+4.8%	+13.6%
2024	+5.6%	-6.9%	+10.2%	+8.4%	+4.5%	+0.5%
2025	-4.2%	-4.4%	+1.6%	+2.7%	+1.2%	+2.6%
2026	+2.5%	+4.5%	+17.5%	+8.5%	+12.0%	+8.1%

7. Trading record

The rebalance log per universe is in `results/xs_FS12_record_<universe>.csv` (names per leg, turnover, leg returns, gross and net L/S).

Universe	Rebalances	Names long / short	Turnover long leg / month	Turnover short leg / month	L/S months positive	Worst L/S month	When
US	163	51 / 50	45%	24%	36%	-23.8%	2026-04
EU	163	24 / 23	47%	25%	37%	-12.2%	2022-03
UK	163	24 / 23	49%	22%	38%	-11.7%	2016-04
DK	163	7 / 6	24%	21%	46%	-18.5%	2020-03
SCANDI	163	13 / 12	32%	25%	44%	-14.7%	2026-01
World	163	98 / 97	47%	25%	39%	-12.9%	2026-04

Current book (signal at 31 August 2026; first 8 names per leg)

Universe	Long: stocks closest to their 52-week low	Short: stocks furthest above it
US	WYNN, APTV, TJX, PNR, PCG, MLM, NRG, LII	SNDK, MU, LITE, MRNA, WDC, STX, DELL, CIEN

Universe	Long: stocks closest to their 52-week low	Short: stocks furthest above it
EU	AD.AS, BPOST.BR, EL.PA, VNA.DE, UMG.AS, BAR.BR, ELISA.HE, KER.WA	KGH.WA, ASML.AS, MT.AS, MTS.MC, TYRES.HE, STMPA.PA, NOKIA.HE, STMMI.MI
UK	AML.L, SN.L, BAG.L, ENT.L, RTO.L, IMB.L, PRN.L, SAFE.L	CWR.L, CMCX.L, SAGA.L, SSIT.L, HAS.L, CCC.L, KLR.L, TRST.L
DK	TRYG.CO, DSV.CO, RBREW.CO, GN.CO, AMBU-B.CO, COLO-B.CO, CARL-B.CO, BAVA.CO	PNDORA.CO, VWS.CO, MAERSK-B.CO, MAERSK-A.CO, JYSK.CO, DEMANT.CO, ISS.CO, DANSKE.CO
SCANDI	ELISA.HE, AZN.ST, TRYG.CO, KEMIRA.HE, DSV.CO, RBREW.CO, KNEBV.HE, GN.CO	TYRES.HE, NOKIA.HE, NESTE.HE, MAERSK-B.CO, QTCOM.HE, MAERSK-A.CO, VWS.CO, ZEAL.CO

8. Risk and factor attribution

JKP 7 themes (US, UK, DK, World) or French Europe 5F + WML (EU, SCANDI); Newey-West t, 6 lags.

Long/short

| Universe | Factor model | Alpha / yr | t(alpha) | R² | Loadings with |t| ≥ 2 (t) | ---|---|---|---|---|---| US | JKP USA 7 themes | -9.6% | -3.52 | 0.72 | momentum -1.53 (-11.0), low_risk +1.00 (+6.6), short_term_reversal +0.65 (+2.5) || EU | French Europe 5F + WML | -5.0% | -1.25 | 0.36 | RMW +0.68 (+2.1), CMA +0.71 (+2.6), WML -1.01 (-7.3) || UK | JKP GBR 7 themes | -6.9% | -2.30 | 0.54 | mkt +0.19 (+3.5), momentum -1.20 (-8.5), low_risk +0.70 (+3.0), short_term_reversal +0.75 (+4.8) || DK | JKP DNK 7 themes | +1.7% | 0.51 | 0.46 | momentum -1.06 (-8.0), low_risk +0.67 (+5.3) || SCANDI | French Europe 5F + WML | +1.2% | 0.31 | 0.16 | WML -0.61 (-3.7) || World | JKP World 7 themes | -11.3% | -3.47 | 0.43 | mkt +0.19 (+3.2), momentum -0.96 (-7.8), low_risk +0.68 (+3.7) |

Long-only

| Universe | Factor model | Alpha / yr | t(alpha) | R² | Loadings with |t| ≥ 2 (t) | ---|---|---|---|---|---| US | JKP USA 7 themes | +1.2% | 0.66 | 0.91 | mkt +0.95 (+21.0), size -0.25 (-3.1), value +0.22 (+2.8), momentum -0.56 (-8.4), short_term_reversal +0.44 (+3.4) || EU | French Europe 5F + WML | +2.3% | 0.81 | 0.78 | Mkt-RF +0.76 (+9.7), WML -0.62 (-7.3) || UK | JKP GBR 7 themes | +6.6% | 2.38 | 0.77 | mkt +0.64 (+13.7), momentum -0.55 (-3.3), short_term_reversal +0.64 (+3.8) || DK | JKP DNK 7 themes | +7.2% | 2.40 | 0.68 | mkt +0.78 (+10.2), momentum -0.57 (-5.6) || SCANDI | French Europe 5F + WML | +9.0% | 2.78 | 0.59 | Mkt-RF +0.66 (+7.1), RMW +0.52 (+2.1), WML -0.46 (-3.8) || World | JKP World 7 themes | +2.1% | 0.76 | 0.81 | mkt +0.82 (+15.7), momentum -0.42 (-5.0) |

9. Statistical verdict

- **Gate:** |t| > 2.87 (2 books × 6 universes, Bonferroni 0.05/12, two-sided).
- **L/S mean:** positive in none of the six; passes in 0 of 6.
- **L/S CAPM alpha:** -16.6% to -6.4%, |t| past the gate in 4 of 6.
- **Long-only:** Sharpe above the equal-weight universe in none of the six; alpha -9.7% to -2.3%, passing in 0 of 6.
- **Probabilistic Sharpe** of the L/S, P(true Sharpe > 0): US 0.00, EU 0.00, UK 0.01, DK 0.05, SCANDI 0.09, World 0.00.
- **Publication decay:** gross L/S -14.9% to -4.1% a year in 2013–19 against -24.9% to -1.9% in 2020–26.

10. What goes wrong, and how to fix it

Method. Diagnose on the design window (2013–2019); apply the same pre-declared fix ladder used for every sort strategy in FS03–FS12; judge on the holdout (Jan 2020 – Aug 2026) with a pooled six-universe paired bootstrap of the Sharpe difference against the previous step. 25 fix trials across FS03–FS12, gate $p < 0.0020$.

10.1 Diagnosis (design window 2013–2019)

Universe	L/S gross / yr	Beta	Beta drag / yr	CAPM alpha (gross)	Costs / yr	Long leg vs EW / yr	EW vs short leg / yr	Turnover / month	Worst months
US	-4.7%	0.12	+1.7%	-6.4%	-1.6%	-1.8%	-2.8%	68%	2018-08 -8%, 2017-10 -8%, 2018-01 -8%
EU	-8.9%	-0.01	-0.1%	-8.8%	-1.7%	-4.1%	-4.8%	72%	2013-04 -9%, 2016-07 -9%, 2016-09 -8%
UK	-14.9%	0.28	+3.7%	-18.6%	-1.7%	-1.5%	-13.4%	70%	2016-04 -12%, 2014-12 -11%, 2016-06 -10%
DK	-8.9%	-0.18	-2.7%	-6.2%	-1.0%	-3.4%	-5.5%	40%	2013-05 -18%, 2018-06 -11%, 2013-09 -10%
SCANDI	-4.1%	-0.03	-0.5%	-3.6%	-1.3%	-1.5%	-2.7%	54%	2013-09 -11%, 2016-11 -10%, 2013-05 -7%
World	-6.2%	0.19	+2.4%	-8.5%	-1.7%	-2.1%	-4.1%	72%	2018-08 -8%, 2017-10 -6%, 2017-08 -6%

On average across the six universes the gross spread was -7.9% a year, of which the market exposure (beta +0.06) contributed +0.7%; the beta-adjusted spread (CAPM alpha) was -8.7%. Costs took 1.5% a year at 63% monthly turnover across both legs. The long leg beat the universe by -2.4% and the short leg lagged it by -5.5% a year, so most of the spread comes from the long side. Holding longer does not change the picture much: the gross spread 2, 3 and 6 months after formation averages +10.8%, +11.0%, +10.1% a year.

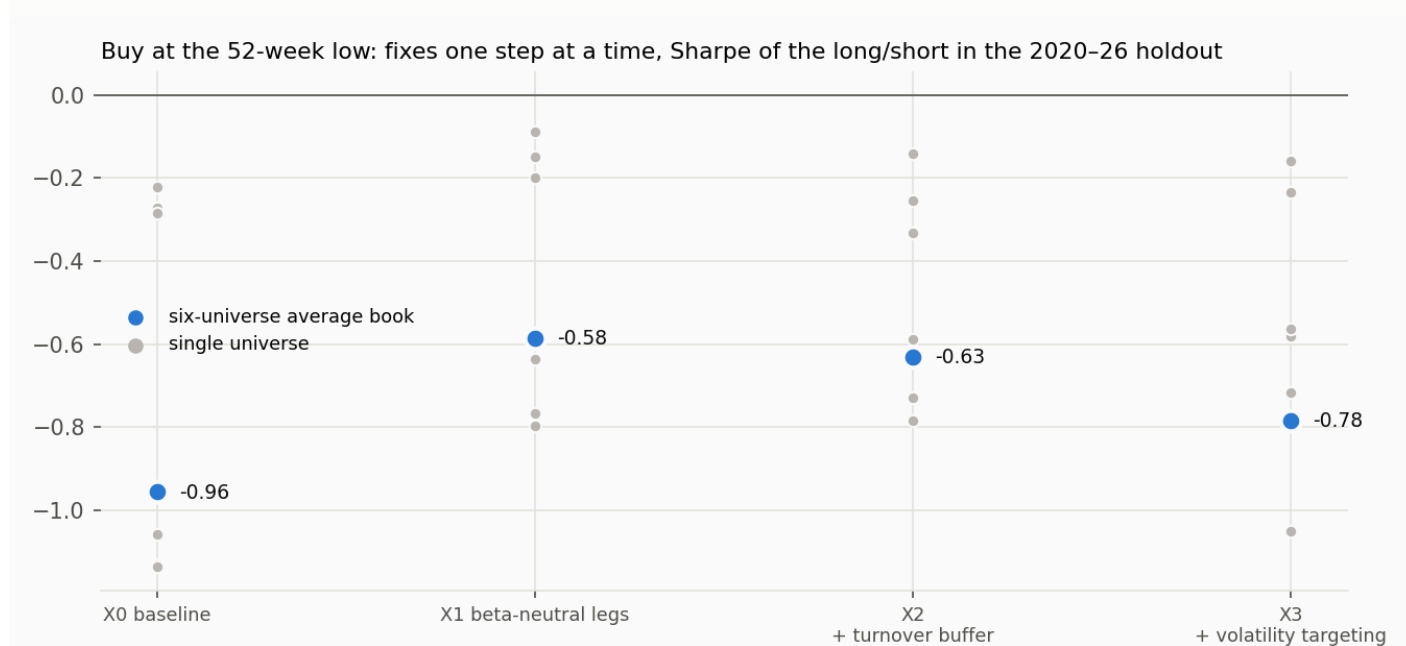
10.2 The fix ladder

X0 baseline → X1 **beta-neutral legs** (each leg scaled by 1/its ex-ante beta) → X2 + **turnover buffer** (enter the extreme 1/q, keep a name until it leaves the extreme 3/q) → X3 + **volatility targeting** (scale the L/S to 10% a year using its trailing 6-month volatility, lagged; Barroso & Santa-Clara 2015; Moreira & Muir 2017).

Step	Design Sharpe (avg)	Δ design (p)	Holdout Sharpe (book)	Δ holdout (p)	Universes improved	Holdout return / yr	Holdout max DD	Holdout alpha vs EW (t)	Costs / yr
X0 baseline	-0.70	–	-0.96	–	–	-12.8%	-64%	-14.9% (-2.6)	1.5%

Step	Design Sharpe (avg)	Δ design (p)	Holdout Sharpe (book)	Δ holdout (p)	Universes improved	Holdout return / yr	Holdout max DD	Holdout alpha vs EW (t)	Costs / yr
X1 beta-neutral legs	-0.53	+0.22 (p 0.046)	-0.58	+0.37 (p 0.053)	6 of 6	-8.0%	-55%	-11.4% (-2.1)	1.7%
X2 + turnover buffer	-0.60	-0.09 (p 0.879)	-0.63	-0.05 (p 0.632)	4 of 6	-6.9%	-46%	-9.7% (-2.1)	0.8%
X3 + volatility targeting	-0.55	+0.11 (p 0.406)	-0.78	-0.15 (p 0.931)	4 of 6	-8.1%	-48%	-10.8% (-2.5)	0.8%

✓ = passes the gate in the holdout. Design Sharpe = average across universes; tests and holdout columns use the six-universe equal-weighted book.



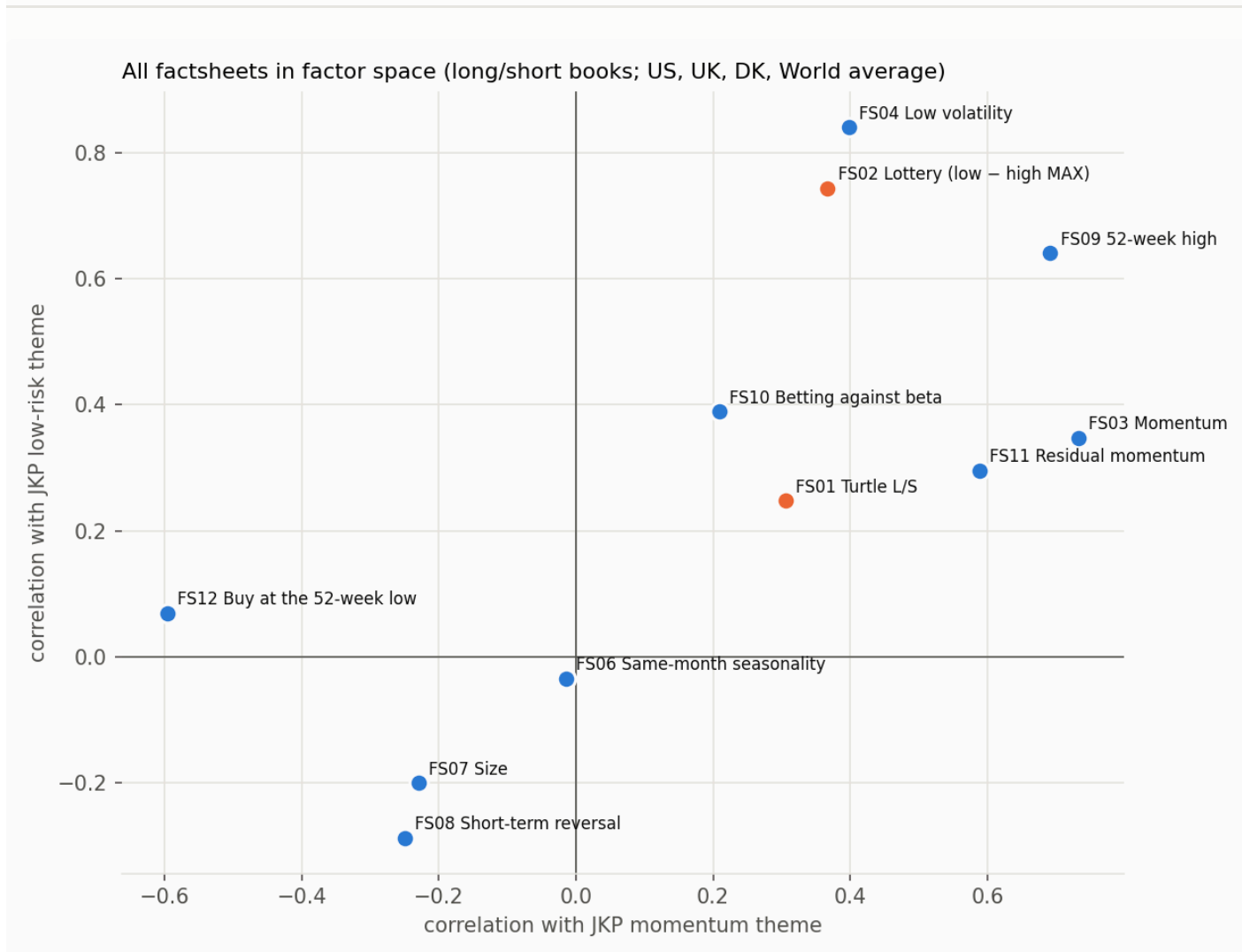
- **X1 beta-neutral legs:** +0.22 design, +0.37 holdout; helps in both windows but does not pass the gate.
- **X2 + turnover buffer:** -0.09 design, -0.05 holdout; hurts in both windows.
- **X3 + volatility targeting:** +0.11 design, -0.15 holdout; helps in one window and hurts in the other: not reliable.

Where it ends: holdout Sharpe -0.96 → -0.78, return -12.8% → -8.1% a year (six-universe book).

11. Classification: where does it fit?

Dimension	Buy at the 52-week low
Series family	Reversal & bottom-fishing
Academic style	Contrarian / bottom-fishing
Signal	Price only (daily total returns)

Dimension	Buy at the 52-week low
Cross-section vs time series	Cross-sectional: stocks ranked against each other each month
Horizon and turnover	One month; 63% of the two legs replaced monthly
Market exposure	L/S beta -0.08 to 0.29
Payoff shape	Treynor–Mazuy γ -2.07 to 2.99 (t -1.3 to 2.0); worst 10% of market months -2.0% to +0.2% a month, best 10% -1.6% to +1.4%
Nearest factor theme	momentum (4 of 6 universes), WML (2 of 6 universes)
Nearest library signals (returns)	Momentum 12-1 (6 universes), Last month return (5 universes), Residual momentum (3 universes)
Economic rationale	Folklore: mean reversion and value. Evidence: momentum and anchoring (George & Hwang 2004) predict continued weakness; losers carry high beta and volatility.
Publication	No academic origin; George & Hwang 2004 as the counter-claim; series A10 (Free Fallin')
Where it fits	In the momentum cluster of the library; fully explained by its neighbours.

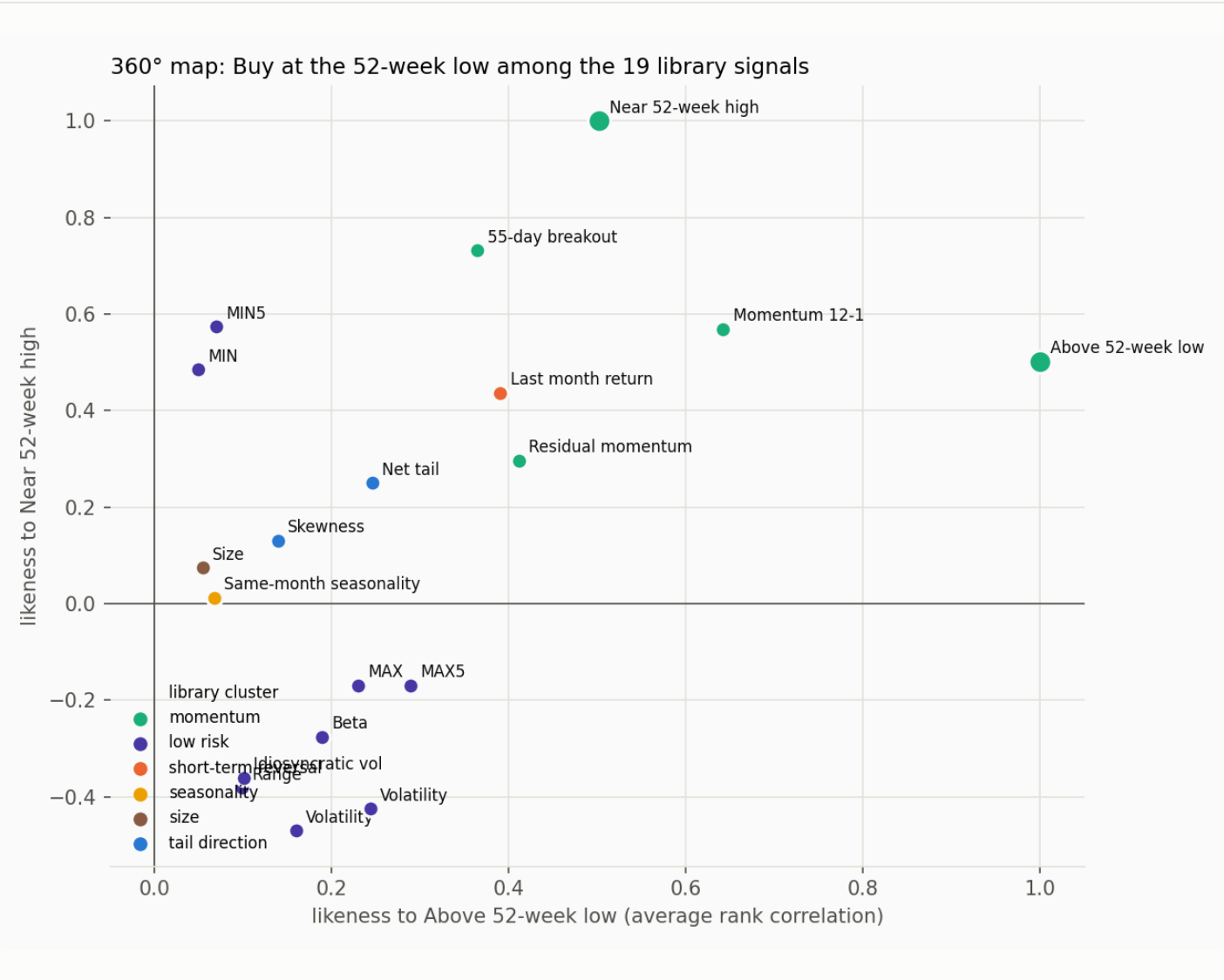


12. 360° view

The signal among the 19 price signals of the shared library (`code/library.py`), on the same universes and months (equal-weighted top minus bottom quantile, gross). Its natural mirror here is **Near 52-week high**.

Signal (top minus bottom, raw sign)	Cluster	Likeness: stocks	Likeness: returns	Raw L/S, avg of 6	CAPM alpha range	Universes alpha t $\geq +2$ / ≤ -2
Above 52-week low (this strategy)	momentum	+1.00	+1.00	+8.8%	+4.4% to +13.8%	4 / 0
Momentum 12-1	momentum	+0.64	+0.65	+7.7%	+3.9% to +17.8%	5 / 0
Near 52-week high (mirror)	momentum	+0.50	+0.46	+2.5%	+8.3% to +18.8%	6 / 0
Last month return	short-term reversal	+0.39	+0.46	+0.9%	-3.2% to +8.4%	0 / 0
Residual momentum	momentum	+0.41	+0.46	+6.1%	+7.3% to +13.0%	5 / 0
55-day breakout	momentum	+0.36	+0.40	+0.7%	+5.6% to +12.6%	3 / 0
Net tail (MAX+MIN)	tail direction	+0.25	+0.30	-1.0%	-5.3% to +4.2%	0 / 0
MAX5	low risk	+0.29	+0.21	+2.6%	-8.5% to +0.4%	0 / 2
MAX (best day)	low risk	+0.23	+0.17	+1.9%	-9.3% to +1.0%	0 / 2
Size	size	+0.05	+0.14	-1.5%	-3.8% to +4.6%	0 / 0
Idiosyncratic vol	low risk	+0.10	+0.12	-1.5%	-10.4% to -0.8%	0 / 3
Skewness	tail direction	+0.14	+0.11	-0.2%	-5.4% to +4.5%	0 / 1
MIN5	low risk	+0.07	+0.11	-2.0%	+2.4% to +13.2%	4 / 0
Volatility (252 days)	low risk	+0.24	+0.08	+5.1%	-10.3% to -0.2%	0 / 1
MIN (worst day)	low risk	+0.05	+0.08	-1.5%	+2.9% to +11.0%	3 / 0
Beta	low risk	+0.19	+0.04	+5.5%	-15.7% to -2.6%	0 / 3
Volatility	low risk	+0.16	+0.04	+3.0%	-10.3% to -3.7%	0 / 5
Range (MAX-MIN)	low risk	+0.10	+0.03	+1.2%	-11.3% to -2.3%	0 / 4

Signal (top minus bottom, raw sign)	Cluster	Likeness: stocks	Likeness: returns	Raw L/S, avg of 6	CAPM alpha range	Universes alpha t $\geq +2$ / ≤ -2
Same-month seasonality	seasonality	+0.07	+0.02	-1.8%	-8.5% to +7.2%	0 / 0



Spanning. Regressing the L/S on its five closest library neighbours (by return correlation) and the market:

| Universe | Alpha after neighbours / yr | t | R² | Neighbours with |t| ≥ 2 | ---|---|---|---|---| | US | +3.0% | 1.51 | 0.83 | Momentum 12-1 +0.63 (+18.2), Last month return +0.21 (+3.7) | | EU | +4.5% | 1.51 | 0.61 | Momentum 12-1 +0.62 (+4.8), Last month return +0.42 (+5.3), 55-day breakout +0.26 (+2.1), Near 52-week high -0.54 (-3.5) | | UK | +1.2% | 0.36 | 0.56 | Momentum 12-1 +0.44 (+3.1), Residual momentum +0.53 (+2.7), Last month return +0.44 (+4.6) | | DK | -3.9% | -1.63 | 0.53 | Momentum 12-1 +0.53 (+6.5), Last month return +0.24 (+2.5) | | SCANDI | +1.4% | 0.54 | 0.49 | Momentum 12-1 +0.39 (+3.5), Last month return +0.36 (+4.5) | | World | -0.5% | -0.21 | 0.73 | Momentum 12-1 +0.85 (+8.5), Last month return +0.47 (+7.5), Near 52-week high -0.61 (-4.9) |

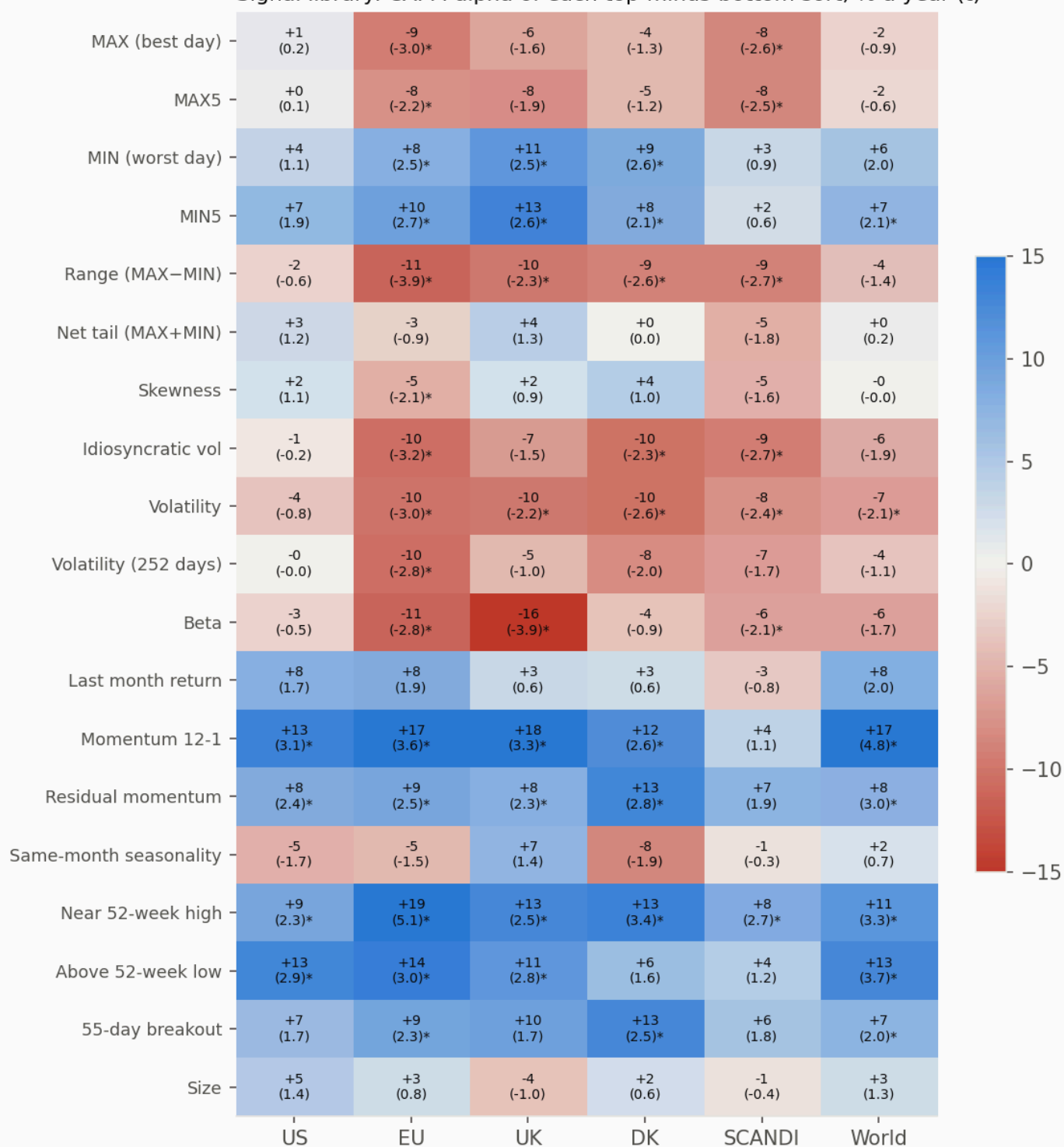
3×3 sort, next-month return a year
(US, EU, UK, SCANDI, World averaged)

low Above 52-week low	10.6%	8.6%	11.2%
middle	12.8%	12.1%	12.3%
high Above 52-week low	14.1%	15.5%	14.7%
	low Near 52-week high	middle	high Near 52-week high

Holding Near 52-week high fixed (down a column), moving from low to high Above 52-week low changes the return by +3.5, +6.9 and +3.6 points a year; holding Above 52-week low fixed (along a row), moving from low to high Near 52-week high changes it by +0.6, -0.5 and +0.6. **Market direction.** Top-minus-bottom return in rising vs falling market months (six-universe average): +7% vs +11% a year. **Horizon:** the gross spread at 1, 2, 3 and 6 months after formation averages +8.8%, +10.8%, +11.0%, +10.1% a year.

The full library, with the CAPM alpha of every signal in every universe:

Signal library: CAPM alpha of each top-minus-bottom sort, % a year (t)



13. Caveats

1. **Residual survivorship.** Point-in-time membership everywhere, but 9–25% of member-quarters outside the US have no price (mostly delisted names); long books are flattered and short books penalised by an unknown amount.
2. **Currency.** Local currency; World sums local-currency returns without conversion.
3. **Equal weighting and flat trading costs.** 10 bp per side for every stock and a size-tiered borrow fee, but no market impact; blue-chip universes only.

4. **Not a registered trial.** Descriptive; the fix ladder is counted inside FS03–FS12 (25 trials).

14. Academic references

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15. Reproduce

`code/data.py` → `signals.py`, `signals2.py`, `library.py` → `xs.py` + `analyse_xs.py` FS12 → `build_xs.py`
FS12. Metrics use Project1 InvestmentLibrary. Licensed and cached prices are not included.